

ActInf Livestream #013.1: "Cybernetic Big Five Theory with the Free Energy

Livestream | 1:51:16

hello everyone and welcome to the active inference Liv stream this is active inference Liv stream 13.1 and it is January 5th 2021 happy New Year everyone welcome to the active inference lab the active inference lab is an experiment in online team communication learning and practice related to active inference feed back you can find us at our website Twitter email YouTube or keybase teams this is a recorded and an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome here and keep in mind video etiquette for live streams for 2021 we are going to be meeting every Tuesday regularly at 7: to 9:00 a.m. pacific time and then we'll plan other special sessions as things come up and for different time zones you can go to this link and see a spreadsheet and here we are in the first week with the first paper and next week we'll be discussing the same one so hello everyone thanks for coming on live and thanks everybody who's watching let's introduce ourselves please give a short introduction or check-in especially if it's your first time on the stream and then pass it to somebody who hasn't spoken so I'm Daniel I'm in Davis and I'll pass it to Colin I'm Colin D young and I'm in Minneapolis Minnesota um I'm a professor in the

psychology department here at the
University of

Minnesota and excited to be here um I'll
pass it on to

Stephen oh hello I'm Stephen I'm based
in Toronto um I'm doing my practice
based PhD at Canary Christ Church
University and uh pleased to be here as
well pass it to

Shannon hey I'm Shannon PRS I'm in South
Dakota so also in the snow but I'm based
at the University of California in
Merced during my PhD also I'll pass it
to

Marco hi I'm Marco um I'm from Holland
and used to work on active entrance in
London but now I'm just um enjoying the
progress from the sidelines

I'll pass it to uh

Ean thank you my name is an I'm from
Russia from Moscow and I interested in
free energy principal and as it related
with systems engineering and I pass it
to uh excuse me what is your name my
friend

Dave let's go to Dave hi Dave Douglas
I'm retired from Information Technology

I live in the Philippines and I did
about equal amounts of graduate level
work in computer science Linguistics and
psychology but have only a lowly

ba cool let's go to blue and then Adam

hi good morning everyone my name is blue
Knight and I am an independent research
consultant based out of New

Mexico and Adam

welcome hi I'm Adam sorry I'm late I
am based out of Bloomington

Indiana uh I'm a postural
researcher cool well welcome everyone
thanks so much for coming on filling out
the Hollywood Square that we have here
for these warm-up questions it'll be
great to hear from everyone just raise
your hand if you have a thought I'll
just put up the first two totally just
raise your hand as you want to add
something first would be what is
something about human personality or
emotions that you would want to see
captured in a good model or
understanding or
education and then also what is
something that you liked or remembered
about the paper and um I think there was
a few memorable parts of the paper just
while people are raising their hand uh I
think just the idea of going from a
statistical understanding of the big
five as principal component Dimensions
to being the outputs of a cybernetic
system is really an interesting topic
so maybe if uh anyone wants to raise
their hand on this one or
Adam or Colin maybe how do you go about
introducing uh the the need for your
kind of paper to an audience like where
do you start out with the interesting
part about personality that you want to
capture in your
model well uh shall I jump in then um
and when you say raise hand do you mean
this little button here or you like
actually just wave at the screen
probably with the button all right well
works that works um so uh one of the

things that I noticed in your summary of the paper uh was something that I often encounter from people working in other scientific fields uh which is a certain degree of skepticism about the big five because of their origin in uh natural language and um and because you know when you see a diagram that's like a hierarchy of personality traits I you know in the figure there I was careful to note that that is not supposed to be a diagram of causal structure in the individual because there's this interesting problem where we use the same visual language for causal models of you know boxes and circles and arrows uh and for statistical models in which we're looking at the relationships among variables but obviously those are not directly translatable and um so the way that I would pose the basic question that's of interest here is uh why do certain traits of people tend to cluster together I think that that's really the core of what I'm interested in and what we're interested in if we're talking about the sources of personality a lot of personality research in the last 40 years or so since the big F emerged as kind of an organizing consensus framework has really been focused on questions of prediction like if we have these assessments of people's personality tendencies what kind of outcomes does it predict for them um also potentially like what are trajectories of uh change and

development of these things over time but I'm interested in a question that is a lot less studied which is where do these patterns come from in the first place um and so I think the way that to think about the kind of statistical element here is simply that there is an observation that if we have a large set of descriptions of uh people's personality and it doesn't have to be from the natural language that was just one good source for a relatively unbiased set of descriptions of people was to go into the dictionary and just pull out the adjectives that could describe people um but we see very similar pattern of factors like the big five if we start with a lot of say existing personality questionnaires that weren't designed to measure the big five you throw them all together you still get the big five um so as long as we have a sufficiently large set of descriptions of personality and we have ratings on them uh they tend to Cluster together in reliable ways and um as long as they are valid you know as long as when we say that somebody's talkative that really does mean that they talk more than other people do and there's a lot of evidence for the basic levels of validity of Personality ratings then the question is why is it that people who are talkative also tend to be uh more sociable also tend to experience more positive emotions also tend to be more physically active also tend to be uh let's say more interested

in excitement and uh
dangerous you know risky Pursuits so
we've got a whole bunch of things that
cluster together and the question is why
and so you know that was a list of
low-level traits that are grouped
together within extra version and that's
just a statistical observation those
people who tend to have more of one of
those things also tend to have more of
another and so I see fundamentally the
question that I'm interested in is what
are the underlying mechanisms uh that
vary from person to person um that drive
these patterns of co-occurrence of
particular
characteristics thank you for that
Shannon thanks for that um really nice
sort of overview introduction um I was
sort of wondering going through this
paper you talk about what are the
differences that are um can be ascribed
to individuals and their individual
personalities that lead to you know
traits like extraversion or what have
you um but what if there is nothing
unique to any individual that gives rise
to a personality trait and I wonder
where does um culture and social
interaction fit into this kind of model
and that's just a question that I'm
interested in throughout the next couple
of hours that we're here
today nice good question anyone I put up
also the third one
blue so something as I was wondering
about like as I was reading the paper
was you know the Big Five personality

and like it's um you know grounded in genetic uh influences right um but like attachment Theory and like there's like a bunch of work that's been done in mice like showing like you know the nurturing of parents and you know whether you're attachment avoidant or attachment anxious or um you know attachment secure that's also something I think would be fun and interesting to um model in in the free energy kind of way like put into into this framework and I think it has like some cool um you know impacts for like team building and and how um you know like whether I don't know I just think it would be it would be interesting and maybe um impactful in in this kind of framework cool and blew to that point I just saw a paper this uh morning from Anna Cuna at all an axle constant the first prior from co-embodiment to co- homeostasis in early life about this transition from like a truly in utero embedded being to different types of physiological dependencies social and cultural so it is interesting to think about all right we have an individual orientation around the big five it's about the individual in relationship to different types of settings but then how do you capture that element of Niche construction or of group dynamics or of these other aspects that are relational I'm I'm just curious Colin or or Adam like where does the uh other how how does that enter into it this idea of the cultural influences in on personality because someone's

personality might be manifested in a really specific setting because of their local context I could just say something about that that is relevant I think to both of the two questions that have just been asked uh which is just to keep in mind that U these personality traits are descriptions of uh you know persistent patterns of people people's behavior and we know that they are genetically influenced but we also know that they're not entirely determined by genetics they are also determined by environmental influences and so the way that I think about this is that you have these uh you know these mechanisms that are allowing people to function and some parameters of those mechanisms must be reasonably persistent to produce these persistent patterns of behavior but the the things that set those parameters and that tweak them if there's change in personality over time can be both genetically programmed and environmentally programmed right so there are things there are events that happen to people for example their relations to their primary caregiver and attachment um that shape what these settings are going to be persistently in life um and so I think there's definitely room for thinking about environmental inputs uh to these systems and shaping of these parameters uh as well as genetic ones cool and Colin this point you raised earlier

about uh graphical models and just drawing a connection among different variables is that a causal model is that a statistical model are those different what if you're talking about causal entropy or you're doing a causal uh you know model like a structural equation modeling or even if it's a linear regression it's going to be interpreted as a causal influence and they'll put in the discussion oh it could be a different way it could just be correlation but it's really hard to disentangle that um yeah

Marco hi thanks um so first of all um I haven't fully read the paper I kind of recently got out of a pandemic um Slumber um but but it sounds very interesting from what I've read and heard so far and as far the first question about what I'd like to see it might be a bit um in the far future but I think ideally um this would all connect sooner or later to the work on neurop phenomenology the kind of Ideal Vision I have is for example now we talking about um genetic prior and environmental inputs but I think the beautiful part about cybernetics is the kind of reflex of action and so Consciousness or phenomenology is basically a mediator or moderator of the way that environmental inputs kind of influence or continuing progression of Personality or um traits and and characteristics and so ideally um I would love to see a well I would love to hear if it's possible to make a

synthesis of for example the work from psychoanalysis Psychotherapy everything that kind of uh explores and navigates these particular mechanisms from a first- person kind of view how to navigate the experience of being engaged in the world and I'm just wondering about Adams or col someone else's thoughts on how to possibly connect that phenomological um description of mechanisms and processes with the more formal um basic models that we're now exploring in this kind of work if that makes sense yeah very nice Adam and then anyone else uh hello can you hear me yep great um so I guess to uh try to address a couple things um so yeah uh Marco just thrust us into like the deepest possible territory like right away um and so I guess I'll briefly comment on that uh Second but the I guess the first thing in terms of um social context and uh Niche construction and the more inter subjective aspects and context um sensitivity of Personality uh active inferential modeling is really well equipped to uh address all of that uh in this paper it was more uh the focus was just kind of exploring what K and I both perceive to be a very rich set of correspondences between between the free energy principle and active inference and uh his theory of cybernetic Big Five Theory but in um modeling work you can uh bring in all of

those details and explore them in their particulars so I'm currently collaborating on a project where we're trying to model things like extroversion within these um social Choice tasks and so how you show up it's going to be a function of you know both the task environment and you and this interaction that's going to evoke a given style pattern but you'll have these um parameters uh these hyper priors are um over the different um aspects of the generative model which through iterated policy selection and uh updating will basically concentrate their probability mass in different ways and result in different parameterizations and then different emergent behavior and what you can see happen is you can get uh these basically attractors form as this process proceeds over time these behavioral attractors where you the concentration of your prior for engaging in the world in a certain way causes you to enact a given pattern which causes you to uh even more update in that direction like your your D Matrix like where am I in this environment uh you know what do you expect to be perceiving your a matrices all the different like components that You' use to try to model the parts of an agent these were all your prior over these all get updated will then cause the pattern of in action so on and so forth and that over time this could cause you to be in this attractor where you have this

enduring stability um how enduring you know it would depend on a variety of things uh and but so I guess that's one so in this process uh the social context who you're interacting with that would make make all the difference in terms of like how the kind of ball rolls down this epigenetic landscape like how you end up like carving out a given pattern for this agent uh with respect to uh neurophenomenology and just the uh actually getting it the richness of experience there's a sense in which all active influential modeling is already in a way um kind of personality modeling it's like you're kind of getting into like um uh these enduring um you these cybernet what they're thinking them as these like cybernetic um parameters for this fundamental process um or just the your ability to describe um the essence of the system uh the it's its normal form description this this maximally encompassing uh description of the modes of where you expect Define the system as a kind of probabilistic density in some kind of multi-dimensional phase space um if you have such a a attracting Network structure that you can describe in a way you could call that the personality of the system um but cybern big five Theory um kind of goes further and and or it it's grounding this in terms of well why should we expect certain modes over others why should we expect this then relates it back to the fundamental

cybernetic cycle of goal Pursuit that any agent has to do um so with respect to neurop phenomenology and it seems like the attempts to get at this within active inference things like uh sophisticated effect of inference the the models of uh metacognition where they're beginning to Model A kinds of mindful processing of reflecting on your um your confidence and and then this influencing your uh you you have having an actual model of your epistemic status you reflect you reflecting on this and this influencing how you sample influencing uh how confident you are in different ways um influencing What policies you end up deploying uh these things to the extent that they allow for um phenomenological richness they could be brought into an agent model and you're beginning to do a connection between neurop phenomenology or a process theory of that would dock of phenomenology and personality modeling so those are some thoughts nice thanks and to add on that Adam because that was really well said it's almost like some of these past papers we've been discussing were the neutral infrastructure or the architecture or the tools to have agents that can have memory or can have affect or can perceive veillance or get worried so we talked that in a very neutral framework like it's a parameter in the model however in the real world as Colin was getting at there's these extremely deep correlations with different outcomes of

systems people who like to text more
also like to video chat more or
something and that might not be
surprising but it these Trends are very
distinct and so we are getting at this
generative model idea personality as a
generator of outcomes in different
niches so Inseparable from the niche
and um it's just an interesting way to
see how the cybernetic big five has come
from a more psychological area so it has
uh that sort of um toe of the neurop
phenomenology and the real correlations
in the world and then there's the
neutral framework that sort of maps onto
that we have Steph and then
Colin yeah I like what's been said about
neurophenomenology and I'd also say that
one of the things that's good about the
this this work with traits is it it's
starting to bring work that's coming
from um what you could say realist
settings codifying it and then finding
the patterns and it's given us more U
material coming the other way we've had
some work around a lot of work with
affordances and niches and then
obviously you've got the models which
start to create Dynamics and show and
then you start to see where these two
things meet but I would also say micro
I've been looking at lot of micr
phenomenology and the actual structuring
of experience and attention and I think
that um what can be interesting is these
traits if these traits can also get
brought down into micr phenomenology and
ways of like how is the

structure of the environment body
mind um happening and and where are
their times because what I think with a
lot of the trait works is it does have a
normative assumption about the type of
Life someone's in around which you then
standardize and then get these patterns
but you do find if you take people into
some other types of space or other types
of attentional State be it a more
grounded State maybe a a performative
state maybe a a space Which is less
Western so to speak less goal
orientated um that might shift so this
might be like this is where that is when
people are at this normative mediated
sort of normative like calculated way of
getting everyone at like the gold
directed Norm for a
Westerner um and then can these
different components to that be sort of
treated and also looked at how do they
vary as we move like not just as the
five but IND you know as as individual
kind of um tap ways to tap into
affordances when do these change more or
less when we bring people into another
type of way of a regime of attention or
another set of affordances which is
outside the normal for our
cultural go driven way of being in the
west okay
Colin and then Marco I just had a couple
of thoughts about phenomenology since
that's been a a theme um recently um
first of all I think one interesting
thing to think about in relation to uh
trait approaches is that when we do

assessments of Personality using self-report questionnaires you know which is not the only way that you can do it of course you can have other people you know rate the personality of somebody and we tend to see very similar results if we use other people's ratings versus self-ratings um although there's also evidence that there's some incremental validity there in other words you know things about yourself that other people don't know so that there's some unique validity to your reports but other people also know things about you that you don't know so there's also some unique validity to other people's reports on your personality but as long as we're talking about the self-reports then that is at base a kind of you know crude phenomenological measure right because you're saying what are you like what is your experience of yourself like are you the kind of person who's anxious are you the kind of person who likes art you know and we get a long list of people's responses on these things and then we uh infer their General patterns of uh behavior from them and so I do think you know one of the things that I've always been interested in is that personality is not just patterns of behavior from a third person perspective it's also patterns of experience from a firstperson perspective and so when we look at the contents of Personality questionnaires you know they involve descriptions of

emotion and types of cognition and patterns of motivation that can all be experienced you know phenomenologically in the first person as well and so we're using this kind of first representation and understanding of ourselves to then try to infer these General patterns of behavior that have validity from an objective third person perspective as well and then just in terms of like drilling down to this kind of micr phenomenology idea we do know things about particular patterns of processing that are associated with different personality traits uh so for example we just published a paper showing that extraversion is associated with the tendency to be influenced by uh reward in an implicit learning context right so you're doing a task where you're making this like discrimination between two very similar options um and you're rewarded more for getting correct responses to one option relative to the other well what you see is that um people's behavior becomes biased toward that option but the degree it becomes biased is actually correlated with extroversion and that's part of how we you know have support for the theory that fundamentally what causes all of these different extroverted behaviors to cohere is sensitivity to reward um and so you can think of that from the third person as like here's a process of you know reward um mapping but you can also think well what is it like from for this individual this is the kind of person

who's going to be more uh perceptive of possibilities for uh rewarding interactions for incentives in their environment so I think you can always sort of flip back and forth between taking the phenomenological and the more U third person perspective on these things cool thanks

Marco yeah uh thanks actually um kind of ties into that because I was kind of wondering if you could share some thoughts on the relation between value systems such as pratic value or affect and epistemic sence and curiosity with this uh model of Personality for example I did read that you said the fundamental components are um the goal the interpretation and the strategy right um and and I would say that that value I mean I would guess that value is more about the interpretation right so that you have some kind of scaler or some other dimension by which you can compare certain current real States and expected States um and tying into that uh I guess it's a separate question but I think I guess I'm going back to the personality and phenomenology because maybe you could share some thoughts and to what extent personality can be seen as an as a natural consequence of the Cy IC imperative where conscience has basically instrumentalized to be the regulator of uh entire mind body system as it were um because I guess I have some image picture of uh reses of reflexive attention right so there must be some cyb active system or set of

systems ecosystem if you will um that's
regulating or controlling the entire
mind body system or large portions of it
um but that in itself then has has to
necessarily buy free energy or back of
inference have to have some kind of um
uh imperative to decide or policies to
pick on how to engage with that bigger
mindbody ecosystem if that makes sense
beautiful thank you Adam and then anyone
else darn it Marco uh that was awesome
yeah was that was epic it was epic Adam
and then Sasha and then we'll leave the
warm-up

so I guess

uh two things Com or a few things coming
to mind

um

so I really like this idea of thinking
of um your Consciousness as a as a kind
of modeling for the sake of cybernetic
control and uh with hopefully you know
some degree of not perfect but some
degree of fidelity for modeling that
which is modeled uh good regulator
theorem to regulate a system you have to
have a decent enough model um
the and so there's a sense in which um
towards this end uh being able to uh
have your modeling process reflect on
itself and model itself uh that that
could be a very useful thing uh if you
get it right

um and a very bad thing if you get it
wrong um the and so related to things
like and it could even be that you know
these parameter Ts for
these they might stripe it across like

multiple levels of abstraction something like curiosity so you could have like a kind of um uh curiosity that has a phenomenology in terms of like the way like a physicist would describe it as like a phenomenological description of behavior that's just like showing the emergent behavior and this looks like a more curious system in the way it's like sussing out state space then you might also get to like uh curiosity in terms of the uh what it feels like to be curious and what those those motivations those free energy gradients they going to try to annihilate those different sources of desire of pull um what that's like

experientially um and so yeah it seems like something like curiosity would be reflected at multiple levels of organization uh maybe even with some uh weird shifts in terms of like the way it shows up across uh levels of analysis um the GU a

few other ideas are coming to mind would be um there seems to be a sense in which if so we're talking about you someone reporting on their experience um you can think of these reports both in terms of the uh perceptual inferences that are uh going into them and then the report is a kind of action as a kind of act of inference and uh each of these aspects you know for in like you could set up a generative model for each of these and you would have for instance a given kind of uh you know matrix to describe the perception like given a given set of

observations what's the state that you then infer and so this could be like some aspects of like so what isn't agent what is a person uh noticing in different circumstances and so this is like what is what is it in particular that is entering into their experience where does their attention go you could also though have like their attention not just in terms of a sort of like uh a salience but a kind of like active curiosity and a a sampling from the environment as a kind of mental act and so this then will influence like what's getting fed into the stream of experience and what it's like to be uh a particular kind of agent particular kind of particular kind of person in a particular kind of situation

um those are some thoughts cool we'll return to that idea about being in the niche for a particular kind of agent we'll do Sasha and then Colin with the last word on the

intro awesome yeah thank you um really great conversation I've already reduced my uncertainty on a lot of the questions I came in with um uh than thank you Colin for explaining that um behavior is not the same thing as personality because that's something I was starting to wonder um so it's yeah relevant that personality uh contains as part of it the self-report of the um individual's experience um and then uh I like what you said about your most recent work that sounds really interesting and kind

of Falls in line with um active
inference uh as reducing uncertainty
about our environment and so it seems
that uh extroverts and introverts have a
different sensitivity to
um uh sens sensitivity to Precision
about their world where extroverts are
more sensitive to reward and um updating
their generative model from the external
inputs like mainly I'm thinking social
but of course could be other things um
so that's an interesting how it fits
into into this model of of active
inference as well and um I guess my last
thought is is there another level up um
here where we can talk about people who
are also sensitive to
Precision about personality tests and um
others Quantified Self
um whether there's a metric for that
kind of like self-reflection part of it
um and if that I think some sometimes
that can seem a bit
like a a self-fulfilling prophecy or
like I know I'm the kind of person who
does this so I expect this personality
test to tell me this um where you're
actually not sensitive to new
information you just want uh kind of
confirmation on what you already think
so but yeah okay interesting stuff Colin
and then we're going to move to the next
section okay so quickly then just a
couple of thoughts on the last couple of
comments
um so first of all you raised something
Sasha that is important which is that
personality tests are certainly fallible

right they're not a great form of measurement they're just the best form of measurement we have for assessing people's personalities and the reason for that is that it's really hard to get at an assessment of people's persistent patterns in all of their you know different aspects of functioning uh with anything other than relying on people to tell you about it right you could try to come up with you know like some complex set of tasks and people come into the lab and do all these things and you videotape them and somehow that's going to you're going to extract all the information about their personality from that by a carefully chosen set of things that you have them go through you know it' be a huge amount of work and you'd be learning less about what they're like in general than just asking them a bunch of questions so you know this is always a perennial problem it's like well there are these difficulties with self-reports people have biases they uh you know they are motivated to think of themselves as being more desirable in various ways than they are from a you know social desirability perspective uh and so what what can you do about that and yeah there are certainly studies of you know the degree to which people are accurate in their personality assessment and like the degree to which if they assess their own personality how much does that agree with other people's assessments of their personality and larger discrepancies can be predictive of things sometimes of

psychological problems um so there's a
you know there's a whole area of
studying that um but I think I like this
idea of people who have curiosity about
their own personalities um and that kind
of leads into something I just wanted to
say generally about personality which is
that definitely uh extroversion is
related to this kind of curiosity about
things that are potentially rewarding
but if we think about curiosity more
generally as a sensitivity to the reward
value of information per se so just
essentially the motivation to improve
the quality of our interpretations of
the world in terms of that uh language
of interpretations and goals and
strategies then we have a there's a
pretty clear sense that Curiosity of
that sort of general form is the core of
one of the big five traits so the
openness to experience and intellect
dimension of the big five is really
focused on people's curiosity uh and
hunger for information essentially
whether that be more about aesthetic and
perceptual information related to
openness to experience or whether it's
related to abstract information related
to to intellectual issues um so that
that seems to be one of the really core
dimensions of people's personalities so
I do think there's a there's a very
clear way to connect that into this idea
of you know foraging for uh for
information and seeking out increased
entropy for the sake of reducing entropy
uh over time you know which is what

learning is essentially so there there
are good connections there
definitely interesting and Adam just one
quick last
note uh so yeah quickly
um guess before we temporarily close off
the phenomenology thread um it's seems
like there's a really uh Rich kind of
biral relationship between uh
personality science an active inference
and that uh with personality science you
actually have this
like like they're all basically these um
uh from the personality modelers I met
they're all these very like integrative
generalists approaching this hairball of
a problem from as many ways as they can
and basically bootstrapping their
themselves to validity and reliability
and so they've come up with uh a really
rich literature of phenomena to populate
the
phenomenology and AC of inference
provides the sort of like places where
you want to kind of dig in deeper onto
like well what is the exact like causal
mechanism so like cybernetic Big Five
theory was probably like best um
impersonality science of getting at
causal processes of like how you're
uh what where are these traits coming
from and what are they actually doing
for the agent so now you can bring an
act of inference and describe agents of
cybernetic systems run simulations uh
generate things that would correspond to
different aspects of Personality like
for instance the dopamine system and

curiosity you now can modify your parameter over policy selection and see what then unfolds with respect to both over Acts or mental acts like attentional shifts um just one final thing before we go to the next part is uh there's some uh in the paper we start to talk about this idea of like a p struct um or like this kind of object that could be used in something like comput computational Psychiatry for um kind of pooling data over multiple multiple people to like come up with um basically a general personality model that you can deploy for different agents um we might want to go into the Weeds on that later um to the extent we can but the thing I want to get at is that uh the one thing to model an AC of inference is we are all personality modelers it's like we are all necessarily like super forecasters trying to predict the future we are all also personality modelers trying to predict each other and ourselves and part of this would be towards this end well what kind of agent do I think I am what do I think my personality is like what do I think is going to be either more situation like what's going to be enduring uh what's going to be you can have personality be enduring but variable so what's going to be uh how variable are these aspects of Me How likely are they to endure these are the questions you have to ask of yourself and other agents in order to govern the interactions in the best of your ability

well said Adam it really just highlights how much people even though we say things like well we don't know even given someone's personality we don't know what they're going to do it's like yeah but you know what language they're going to speak and so in the broad scheme of things you've actually reduced your uncertainty quite a lot and if you're not 99.9% aligned with that agent who's like you in the niche sharing an interface with you that conflict or adversarial relationship or just risk surface is going to spiral and so it's all about this interface yeah Marco if you want to go ahead yeah thanks U sorry I I didn't get get that far in the paper but that sound really interesting I just briefly skimmed the section on it so so it sounds like the P structure is basically allowing for a heris contextual dependency so they have the generic generative models of your being and the and the P structure kind of uh links the current context to the generic uh kind of models and that allows for the variation over different situations I mean I'm emphasizing theistic nature of it because personality basically can be seen as a very useful theistic for how to act in the world does that make sense or should I just read more in the paper no I think that's really a great uh connection there and a setup I think in figure two or when we see the graphical model maybe let's think about where this structure would play in and then

emphasize its role as a statistical construct that reduces our uncertainty about regularities in the world just like every other kind of statistical modeling framework does um and then I think we'll be able to come back to that P stru question but what a fun uh warm-up

section good times so in today we did the introductions and warm-up and then we'll have the sections of this paper which maybe if you're listening you've read or not and that is the integrating cybernetic Big Five Theory with free energy principle a new strategy for modeling personalities as complex systems by saffron and D young both of whom are on this call so in this set of slides we do have a bunch of uh the abstract road map some other stuff but I think there are so many interesting threads we've raised let's just try to walk through all of them um in no specific order but really think about what topics do we want to highlight for next week when we talk about the same paper and how will we get deeper into some of these threads that we've raised today in the discussion so here's the ases and claims slide and Adam or Colin um what was just your intention with the paper and after doing it where do you see the next implications of the paper or the next research directions for what you kind of want to do in that direction and if that verges into some of the figures just me mention it and I'll go to that

figure um well I think the our basic motivation for doing it was just that we had already been talking together about topics of mutual interest and and the opportunity to write this chapter came up and I thought oh this would be a really fun thing to uh a good excuse to do something with Adam and kind of get our thoughts together and you know actually written down and out there in the world um and so you know I think he and I probably have slightly different incentives for where this is going next um he's more interested in the process processes of modeling um whereas I'm more interested in you know generating knowledge about the specific mechanisms and uh you know brain processes that underly different personality traits and risks for different types of psychopathology that are tied to different personality traits um but um yeah I think I mean to me this is just a really nice way to kind of connect U my thinking about personality with this uh what seems to me to be a really generative perspective with the whole Fe framework and what's happening with that in science in general I'm sure Adam has some other thoughts about that too yeah go for it Adam but that was very interesting Colin thanks oh I mean for me that was primarily the thing like col and I were already kind of going into the weeds together and having fun with it and so it's then uh

basically uh uh iarpa brought together
this meeting of personality modelers to
create this book um modeling persons and
situations um basically seeing the
extent to which personality science
could do uh predictive modeling of
Agents um basically they want to recapit
the plot of westw world as far as I can
tell um so I'm actually not joking but
except I am uh but yeah so my primarily
uh uh interest would be just uh what the
heck our persons uh and uh what can they
be and how can we understand them and uh
who am I and what do I
want yeah and also one thing we haven't
raised is who or what is a person
there's personality as being the
statistic iCal regularities of agent in
Niche so that's just uh like a person in
an exoskeleton you could have some sort
of model for that or a person with other
types of constraints on or something
like that but then there's the idea of
like what is a person legally what is a
person culturally and all these ideas
are coming into play with modeling legal
persons and physical persons and regul
statistical regularities that include
our digital affordances all of that is
becoming integrated our digital Persona
is that included in these uh emotive
models there's just a lot of really
interesting stuff to think about there
so then just copied on the screen for
people who are looking is the abstract
you can pause the video there and here
is the road map then we'll have the
figures and then there's just one kind

of interesting general question from an audience but in the road map just if you have any thoughts on the layout or the authors could give a point if they wanted to about why they arranged it the way they arranged it or which parts they thought were important if somebody was just going to read one section or any thoughts on that otherwise we can just go to the figures but okay let me know if you want to return to this otherwise it's just the headers of the paper all right so here's figure one and Colin this really returns to your point from earlier about the graphical models what do Connections in graphical models represent and that's very related to a general question of like what do variables represent principal component one what does that represent what English word are we going to put on that you know $MX + b$ linear model what are we going to call x what is the slope what is that going to be called and then how does that play out graphically whether in an equation or whether in a drawing like this so I would love to hear your thoughts and then anyone else can raise their hand on this one okay so I made the basic distinction earlier between a uh a St a statistical model of covariance like a structural equation model uh I think it's really important not to call structural equation models causal models because they're not that drives me crazy when people refer to causal modeling and all they're doing is a set of regression

paths right what that is fundamentally is trying to U you know recreate patterns of covariance in data and if you have special kinds of data like with temporal priority or where some variable has actually been manipulated uh then you can actually interpret um you know causal effects from covariant structure but uh just in and of itself there's nothing inherently causal about structural equation modeling or regression um this model here however is different this is actually intended as a conceptual causal model so these are not just looking at correlations between different variables in a population this is thinking about the actual causal processes that produce the functions of a of an individual agent specifically a human being um and so what this is this is just sort of like a way to encapsulate an overview of the way that my theory uh defines personality traits and characteristic adaptations which are the two elements of exhaustive elements of Personality uh in the theory and so the idea is that you know we know from Decades of research that both genetics and the environment contribute to uh people's personality traits and the uh idea is that personality traits are patterns of behavior that's why they have their own separate box there there's sometimes a tendency in personality psychology to say well what the trait really is is some underlying thing that causes people's behavior um but I think that

that leads to confusion because what we actually measure when we measure personality traits is a description of regularities in the person's behavior and experience uh so I think it's important to keep that clear but then the the middle box there the relatively stable parameters of cybernetic mechanisms is trying to say something about where those patterns of regularity come from and so the personality traits are not some underlying causal process that generates personality they are just the patterns now they themselves do have Downstream causal effects however because let's say I'm the kind of person who talks all the time well that's going to have consequences on the world around me right and that will maybe lead me to develop different patterns of adaptation to my environment to have different outcomes on the people around me it's going to lead to different outcomes and different adaptations so that's the final box there um and then there just you know there's identification of feedback loops here so the idea that you're generally speaking the flow is that your basic personality traits caused by these basic settings of parameters that have a strong genetic component to them influence how we adapt and how we you know what we pick up on and what habits we develop in any particular situations uh habits is a good synonym for characteristic adaptations um but sometimes there can actually be feedback so let's say for

example that I'm an introvert but uh for whatever reason I get into a job that has a an important component is selling things right I get into sales somehow so I'm going to have to develop at least in that context the ability to act like an extrovert if I'm going to be successful in my job and maybe I get good at that right so I've got a characteristic adaptation to my particular Niche which is that at least in this context I'm good at acting like an extrovert but this acknowledges that there's the possibility for that to feed back and actually make me more sensitive to incentive reward words in general and so I may actually find that I'm not just more extroverted in this particular job context after five years of doing this maybe I'm more extroverted in my life as a whole in all these other contexts um so that's the one of those feedback loops and then so you see that there's a the environment can directly affect the kind of ways in which we adapt to our environment obviously but the environment um but genetics any effect of genetics on our specific adaptations must feed through the effects of genetics on these General mechanisms these parameters that underly our functioning um and so you know if anybody has any other questions about that diagram I'm happy to respond that was a great explanation and just to um emphasize one point there and then we'll go to Stephen you said that the characteristic adaptations in

life outcomes are exhaustive and so um
that really is like we can partition
them into the adaptations or the things
that the agent carries with them the
synapses that have changed or the scars
they bear but basically what is not
their surrounding and the life outcome
is the niche so then there's this stigma
G where the agent is doing Niche
construction so this is from a different
nonactive inference non- Niche
construction framework but finding
resonances and finding uh concordances
here because that's kind of what we talk
about like the learning of agents and
then the way that influences their Niche
and then you had this idea of acting as
if like the example with the person
who's um being forced by their Niche to
implement uh I'm a kind of person who
says hello in this way or things like
that and then that also relates to what
Steph was saying earlier about drama and
about performance so maybe Stephen do
you have a thought on
that yeah I mean the the the thing I
like about what was mentioned here is
how personality traits can be adapted to
context like someone's A salesperson I
thought was a useful example and then
how that obvious starts to then change
your whole broader way of seeing things
and one area I think might be
interesting and I've been there's the
work of Harry Hest who's revisiting this
the work of Roger Barker from the 60s we
did a lot of work on Behavior settings
so he looked at a whole load of children

across the US and he looked at how their behavior was when they were in a school or in the cafe or different places he found the difference in

Behavior between settings for the same child was bigger than the difference in

Behavior between personalities of children I the same child was changing

its Behavior radically based on the settings um and that that work kind of got lost but Harry Heft has sort of been really pushing that again I I think that can tie into this is maybe there's personality

traits I suppose you you know how much stability there is but could Behavior settings be kind of like that kind of more Dynamic version personality traits and I think active

influence makes that more possible to even have that conversation rather than the more sort of static

psychometric data which is kind of topped down and kind of you just have to accept it feels like this is M you can get in there and play with it a bit more from different

directions thanks for assuring that

Stephen we'll go Colin and then anyone else who raises their hand yeah so that's a really a really good point

Stephen which is that

uh there is there are differences that people have to specific situations that are distinct from their broad patterns of consistency and personality and that's a long running debate in

personality psychology going back to the

to the late 60s um and the resolution to it basically is that yes there are these relatively stable ways in which people respond to specific situations and that's what I'm trying to capture with the idea of characteristic adaptations but then if you go across a bunch of character a bunch of situations and you take the person's average there are also persistent averages that differ from person to person and that would correspond to the trait so you can have both of these things in the same set of data um andh will fleeen is a personality psychologist who has really been instrumental in collecting the data to show this very basic feature of what are we measuring when we measure a personality trait with something like a questionnaire where we're just asking somebody for this sort of general impression of themselves and so what he's done is used experience sampling methods where you like you have somebody's smartphone and it you know pings them like five times a day and asks them like how have you been acting for the last you know half an hour or for the last hour and then you get them to rate their behavior on a set of uh you know on a set of descriptions or categories what you see is that uh as you said people have as much or even more variability within their own behavior as there is variability in terms of their average behavior from person to person and yet nonetheless those averages are quite stable so if

you do this for one week you can get a whole you'll get a he calls them density distributions right you basically just get a you know and they're usually fairly normally shaped um and you do that for one week you get this whole pattern so sometimes the person's acting very introverted sometimes they're acting very extroverted you know at ranges um but there's there's a peak and that Peak is very stable if you do it again in another week so the person goes through another week of behavior and again they're varying very I in their behavior but if you look at their average level of that particular Behavior it's quite stable over time and so and then he's also shown that that correlates quite highly with the ratings that we get from these General personality questionnaires so the questionnaires really are tapping into this sort of General average tendency of specific behavior even though of course the behavior itself varies widely from situation to situation and over time both due to situational affordances and due to whatever is going on internally in terms of the goals that the person is pursuing at that moment awesome let's go Marco then

Adam thanks it was actually really interesting I apologize I'm going to add another interesting thing to connect to so so so someone mentioned Steph mentioned the uh the kids in the classrooms and the context and I think development might be also very

interesting here because um personality might be relatively stable across the lifetime but but I think hopefully we can agree that personality also enriches um and in in that process of enrichment and I would also say empowerment um there's I think you could say a shift between the balance of endogenous and exogenous influence on an agent's behavioral adaptation or to its to its niches um and so there's a there's a change in the scope in which the personality uh has a relatively large influence on the way they adapt and I guess I guess I'm just asking about thoughts on how you might picture that kind of process of how personality might actually in the beginning um largely concern itself with how to engage with the world such that it might internalize that which it normally co-ops from the environment so normally offload a lot of stuff to behaviorally adapt which might explain the the variation due to environments but then over time as it does so it internalizes certain heris certain patterns certain policies and learn from environments to kind of enrich its personality and then extend the scope uh of of influence from their personality um yeah I don't know if that makes sense but yeah cool thought Adam and than anyone else uh so few thoughts from before and then um some thoughts on Marco's thoughts um so the flee and description of uh personality as a kind of a density uh distribution um I think and the issue

of um the stability of personality and the variance of it um with within individuals across context which is part of the reason the uh book that was this chapter was part of was called modeling persons and situations but uh taking these ideas together uh we're kind of in the same territory uh that of these fundamental questions in the free energy principle of like what is the ontological status of a non-equilibrium study State distribution bounded by a Markov blanket um what's the granularity over which we can find this this thing that we're identifying so it's you're uh and so um col will be able to speak more to this but this idea of like there are ways of describing that uh your degree of variability across context is itself something that can be stable and this could be itself part of your personality um to uh now kind of loop back uh to what you were just describing

Marco

um I really like that idea um so it's like you're you're starting out in a situation where through this interaction between you and this context um that you're entering into in certain ways based on your own expectations and the expectations of people in the environment and and the structures that are there built up from this there's a certain stability and then you Bo you uh this stability um ends up shaping you um both just as as a consequence of like this is the thing for you to predict but

also in the active inferential sense and
as you're saying empowerment this is
what's going to help me get better
purchase I'm reducing my uncertainty
with respect to realizing the value of
uh existing and so this um idea your
personality being kind of um
internalization of structure and maybe
at a certain stage almost like a kind of
amortized inference um in terms
of I got from you
uh yeah I really like that idea a lot
cool Adam

Shannon you're muted ch

thank you um going on this developmental
thread and um the paper that you brought
up this morning Daniel and um Anna's
newpaper on the the first priors when
you're in ruo as a baby I haven't read
it yet but

um the idea that like when you're in a
neonate in a body you're literally
relying on someone
else's interceptive States and
biological states to regulate everything
about your inceptive in biological
States and then you're born and you're
still kind of relying on um that
caregivers inceptive States or their
behavior when they're holding you or
coming to comfort

you um

this I think could relate to what Marco
was saying about how your personality de
velops so first if you have if a
personality is something like a
collection of effective States and
behaviors

and an infant is completely sort of
reliant on the environment and the
social environment to regulate its own
personalities and as you grow through
toddlerhood you slowly are able to
control this yourself like you can
maintain your own body budget a little
bit more um learn how to how to get your
own blanket if you're cold or like self
soothe and then you grow older and maybe
you're taught how to make music and
learning that this is a new way that you
can explore your personality or create
it um

and I don't study development at all so
sorry if this is wondering maybe Sasha
can pick up on it but I I feel like this
can kind of connect if instead of
looking or not instead of but in
addition to looking at averages but
looking at like a Time series of
personality development over time and
like you could do at any point in time
or at any range of time some sort of um
like what would be the the active
interests model of your life in infancy
in toddlerhood in you know Rowdy teenage
years and as adult and you'll have you
you'll find these like steady
trajectories at the same time that a bit
R Hope some can pick that up let me
connect that to a few other points So
Adam raised this question of what is the
level of the thing or how do we identify
things from empirical data specifically
so that we can use it in active
inference type or free energy principle
type models do we use renormalization

group or principal component analysis or some type of adjacency Matrix some type of Markov blanket what are those things that we're actually going to be modeling and that reminds me a lot of a theory called an actor Network Theory by Bruno Lor and then also on the ant wavelength Colin was mentioning how there's so much variation Within individuals but also stable differences between and that reminded me a lot of my graduate school work with ant colonies where day-to-day the variation across all the ant colonies was extreme if it was raining they would do one thing if it was very dry versus if it had rained the day before those were massive determiners of colony behavior however across days and even across years there were stable markers of colony behavior and there were characteristic ways in which they responded to the environment so it is possible to have a type of measurement and understanding of a system which my adviser established through long-term observation where you can actually model this development through time as well as within a population which doesn't erase individual differences it actually is the framework by which individual differences arise and then Shannon is so beautiful with music being introduced like a thread of personality and then that can be used to explore these regularities Adam then Stephen hi um what was the paper that people are mentioning uh about like

first priors I'll post it in this jitsy
and we'll put it in the YouTube chat but
let's continue without worrying about
this paper
okay go on oh
sorry should I go do you want to can I I
just I was just going to put in one
thing maybe I just um
because I like when it mentions here
relatively stable parameters of cyber
cybernetic mechanisms sort of those kind
of Dynamics and I think that links also
to this idea of how that might change
during our life and thinking about our
lifetime and one thing that I'm very
interested in is when trauma comes
into play PTSD and other types of
situations and you know it's um it can
really shift the state of someone and
poly vagal Theory kind of talks to that
where we've kind of got this sort of
social co-regulation that we have as
humans but we also have our kind of
Primal kind of fight or flight or gut
sort of sort of fear um responses and
how those things are in a way they
another type of like patent Behavior
which becomes incredibly stable too
stable like you can shut down or you
might get into rage but then over the
period of the day or life the challeng
is they may it may be unregulated about
when they appear so someone might sort
of be a certain way and then they flash
into rage you know and then they come
come back into and their personality is
radically different and I think this is
so there they might be interesting to

bring in that that that as a as a kind
of a what happens when that relatively
stable State itself flips you know um so
the and it so your your your mechanism
might not be cognitive but it might be
in that kind of um polyal system yes the
body influencing phase transitions in
personality or behavior Adam and then
Colin and then

Marco we muted

ad thanks um yes I think uh Colin should
probably speak to what was just brought
up in terms of like what actually
happens with people um but in terms of
uh like a high level of abstraction um
so yeah it seems like there's this sense
in which you can think of Personality as
a kind of core screening but it's and a
kind of renormalization and um in terms
of it existing as a phenomenon and it's
a and epistemically it's a kind of core
straining um afforded by this
renormalization um the this kind of
throwing away a variability as you form
these higher level attracting States um
that you can uh ignore them uh with
within the scope of a certain type of
modeling that you're engaging in with
respect to having certain objectives of
trying to predict certain things um
whether you're a personality modeler
trying to predict someone else or trying
to predict yourself or whatever um
the So within this um looking for this
like
optimal level of epistemic grip in terms
of like the granularity of your
description um

this there you could describe these cybernetic control parameters as being uh part of the causal process of what's responsible and then if you take the system and you perturb it there's going to be like a range where like the order parameter changes and you can get these nonlinear maybe phase shifts where the um you're now in a different regime and it may look like a very different kind of person because the personality structure as this stable attractor Network um can then become destabilized um and so that would be like a high level in colge probably speak to like what actually happens with people cool yeah call and go for it so you get these phase shifts in Behavior but they're often due to persistent sensitivities in the underlying mechanisms right so let's take um well so uh PTSD okay um what you have there essentially is some kind of traumatic event uh is an intervention that resets the sensitivity of the threat systems and so the that corresponds quite well to the big five dimension that's labeled as neuroticism which is an old-fashioned word but in this context basically just means sensitivity to all different kinds of negative emotion um and so obviously that's heightened in PTSD and what that means fundamentally is that you've gone through some kind of traumatic event that led to your your it wasn't just learning about a specific event right what it did was that it reset the sensitivity of your threat system as a

whole so now you are perceiving threats and reacting with defensive negative emotion to much more minor situations than you would have previously um and of course even in terms of risk for PTSD we know that people who are higher in threat sensitivity to begin with are more likely to develop PTSD so it's not like it comes out of nowhere typically but nonetheless there obviously this change um and there was a recent really nice meta analysis of Personality change following uh therapy and different types of psychotherapy whether it's pharmaceutical or talk therapy or cognitive behavioral therapy all of these lead to changes in personality and the trait that changes the most as you might expect is reductions in neuroticism right so uh what we're talking about here is the ability to shift these underlying parameters that are not constantly manifest in the same way so when we say that somebody's high in trait anxiety for example we don't mean that they're experiencing anxiety at every minute of every day at a high level right we just mean that they're more likely to be anxious more intensely in more situations than somebody who is low in trait anxiety but nonetheless then if we really want to understand the Dynamics of the system we need to understand those phase shifts from like feelings of safety to feelings of anxiety at any given moment and I that's what you're talking about but in order

to understand how that works for a given agent we also need to understand these persistent patterns of the sensitivities that lead them to have these shifts in different situations and to different degrees than other agents very cool thank you Marco then blue thanks um yeah this is really interesting so um trying to tie back to the earlier talk about um the role of an effect to this aspect of trauma so I think it's mentioned before by the two authors that that um we could uh validly see these complex systems as composed of uh different micro personalities distributed over scale and levels um and and similarly for affect in the sense written about by KAS per has and others so um Adam safron Adam also mentioned renormalization right so so I think that ties in also to a lot of different things like Stig or equilibration and I think perhaps it's it's a had a thought or don't want to share is maybe maybe um the clustering phenomenon of personalities might be a natural consequence of something like renormalization or equilibration if there already is a certain heterogeneity if there already is a certain kind of a symmetry in certain factors certain basic mechanisms of the CTIC uh systems shed throughout uh then that naturally should cluster because then they are more likely to cooperate if it remains heterogeneous and disordered or not co-organized uh then it's inefficient but then you have this thing with trauma

which I think would disrupt that process right because if you go back to the component you pointed out interpretation goals and strategy then that's basically the key things about a trauma right you're in a certain situation that just demands an extreme change in your interpretational beliefs or models your goals for the particular situation and the extreme pressure to have the right strategy to get out of it and I think maybe the the the the side effect right the byproduct of that momentary extremeness is that it basically expands its influence and so it it becomes hypersensitive by nature and so uh the ongoing the always ongoing process of equilibration of everything co-regulating um becomes disrupted by that one participant the traumatic complex um if that makes sense so instead of a lot of different not these kind of equilibrating with each other kind of polycentric it becomes a monocentric equilibration if that makes sense it's AIT rambly lots of very interesting points thank you blue and then Adam so just to kind of take the trauma aspect maybe to the other side of the spectrum and and just in thinking about um you know like our own perception of our self right like so how we hold on to our own like person person ality versus is it you know congruent with what others think of us so I've thinking you know as how does this really relate to um meditation right like so in in meditation there's like the idea of

getting rid of the aita right like so
where like that's the thing that builds
the ego or like the construct of eye so
as we like eliminate this like construct
of eye and also like staying in the
moment right like we're not necessarily
trying to build an accurate predictive
model in in a meditation state so so
when we have this like it's like this
dissolution of of like free energy or or
or what is happening like in in the
meditative State um that that can also
bring about a drastic personality shift
I think as we get get rid of the eye and
learn to be non-reactive as opposed to
like traumatic reaction that's nonlinear
acceleration awesome Adam and then Marco
then Colin and then anyone else who
wants to raise your hand
hi um so I I just try to address a
couple of the threads
um come to meditation uh at the end I
think but the
um it seems
like a lot of the um principles of
personality
uh in terms of their generality uh could
be really interestingly modeled um
actually using things like an colonies
uh like in Daniel's work um and you
might even find interestingly like even
similar mechanisms at play so like in a
recent paper starting to do active
inferential modeling of and Colony uh
Behavior Uh decisions and foraging um
Daniel mentions like you know there's
like the was it the foragers have more
dopamine

so you could see things like um the functional significance of dopamine actually as it's way of parameterizing a cybernetic system as being potentially similar although potentially switching across scales where in the individual ant and maybe even in in the personality of the colony as a whole and uh there's would be actually this interesting interplay between the personalities of individual ants and the personalities of colonies and this interplay which might not be an exactly a one for one might actually even be a kind of um uh complementarity where certain characteristics being present on um this broader level this this more encompassing marov blanket where the individuals are part of a superorganism or just the organism um from the perspective of natural selection because the colonies that which is selected uh as Daniel says um but so the person out of the colony let's say um would then interface and interact with the personality of the uh the individuals of which it's composed and uh potentially in an in training way but potentially not um to loop around um you might even be able to uh at the level of of colony talk about things like the aect of tendencies that it's you know it's what is its what is the affect the veillance the emotions experienced by the colonies is a whole in a given instance what is its um uh General mood and then what are its patterns of mood across different

situations it's and you might now we're kind of going into what we more call personality territory and you could even think of things like um to kind of bring in some of like Marco's idea of the nature of trauma and this like dynamic system sense like you could traumatize The Colony and does it then at some point like there you would expect some sort of like elasticity or um what's the term like Hyer I don't know not that but it would be you perturbate and then it it would have some sort of adapt potential to come back from the peration because that's what we're trying to do is H keep existing in the way we are and so some adaptive potential but you push it too far uh and then you might not get back this um adaptive attracting structure you might get stuck in a different regime um and you could think of the same things uh similar things this could be a way of describing things that happen within the within uh individuals like so there's um you can think of that maybe on multiple levels of abstraction but stop for now thanks yep I just switched the slide to figure two so Colin and then Marco go for it and maybe we can introduce the p struct idea or just how does active inference specifically help us address some of these questions but go ahead call in and then Marco Okay well what I'm my thoughts are about a

couple of the last comments so they're s
more related to the to the previous
slide and then go ahead then it's fine
um so in terms of both this discussion
of what's going on in trauma and what's
going on in meditation I think one of
the interesting things to think about
again is the distinction between the
personality traits and the
characteristic adaptations um because in
uh in both there's the capacity for both
of them to be shifted so uh Marco was
talking about the way in which in a
traumatic event uh you basically
pre-existing goals and interpretations
and strategies are dramatically
invalidated and and that uh you know
then means that the system doesn't know
how to pursue its goals in fact it may
not even know what its goals should be
and obviously that's extremely
disruptive and disregulated and so one
of the challenges in the immediate of
course is to figure out uh what you
should be doing what strategies you
should be using but then there's also
this capacity for that traumatic event
uh to shift the underlying sensitivities
of the system to what is perceived as
disruptive in future right to sensitize
to to things that disrupt um and that is
you know obviously an adapted response
to say like if you're in a in a
dangerous environment you should become
more sensitive to uh to danger and the
traumatic event just basically taught
you that the world was a much more
dangerous place than you thought that's

part of what is uh tragic about it right
because you have this persistent sense
that the world potentially is much is
much more dangerous even than it is or
that you should experience it as being
but then you have the same kinds of
Dynamics going on in um in meditation
where trying to reduce your reactivity
that's like trying to shift some of
these basic parameters of the system so
that you you know react less easily
you're disrupted less easily and then
the kind of non eye idea the recognition
of the the elusory nature of the eye
what that's about is basically not being
overly attached to your uh model of
yourself right because the the model is
a model it's not yourself so if you are
overly attached to your model then you
have you're overly attached to your
conscious understanding of what your
goals and your strategies are and what
you're like and meditation is basically
trying to get you to recognize that
those ideas that you have a they're just
a they're just your model they're not
actually the way that you are and B
they're contingent on your situation and
so you should be willing to let them go
as your situation changes and as things
evolve and so it's essentially creating
both a pattern of Greater flexibility in
the capacity to adapt uh but then also
the attempt to uh downregulate the
negative reaction to disruption right
because if you have more confidence in
your ability to adapt then you should uh
be less defensively reactive to

disruptions because you can be confident that you'll simply adapt to whatever the the new situation is thanks Marco yeah thanks um yeah just just piggybacking on that because that was really nice um so so so my thoughts on meditation and very good and important question I think um so so going back to the image of having all these different distributed micr personalities or sub personalities I think meditation can be seen as allowing them to manifest and arise as it often is called and manifest their the Expressions which are always going on to also reach as a were the global workspace or Consciousness or the uh conscious attention and so what you're doing then effectively is broadening the collective of sub agents or subpersonalities or subsystems such that that equilibrative process improves because then you have a bigger scope a bigger domain of possible inputs to an ongoing process of inference um and importantly this is then of course without need for explicit modeling you're don't need to explicitly label a certain um rationally or verbally explicated belief right it's an exercise in in in the in the deepest sense of it uh importantly it's it's it's effectively a reflexive Niche construction because meditation is systematically to basically dis attenuate to yeah to attenuate external inputs you sit in a very neutral position you you're in a calm environment external input is minimized

and so all the inputs well it augments
the inputs that is endogenous coming
from within which of course um is then
going to translate or transfer to normal
waking States because those are all the
endogenous or it concerns all the
possible indulgent influences on your
ongoing
cybernetics and so it's it's it's a
reflective shift right and yeah I don't
know what I was going to say uh yeah so
it's basically just a sort of game so
from what I have seen my limit
experience with meditation and reading
upon it it's it's all systematic
cybernetic games it puts you in a
situation that that forces you to train
the process that underlies sonetics
concerning the purely reflexive which is
of course beautiful because there's
almost no opportunity for that it's an
extremely exclusive state of being that
exclusively uh forces you to attend to
these internal processes it's kind of the
sbics of the sbics if that makes sense
great points there it's why so much
meditation begins with thankfulness and
just to give actually one behavioral
example um let's think about the example
of a circus elephant where when it's
young it has a heavy chain and it can't
break away and then when it's older a
simple thin rope can keep it connected
so at a first pass one might say um well
why doesn't it just break away why
doesn't it engage in a policy of
breaking away and the habits the E the
priors over the policy the field of

affordances may not even include that as an option so if certain types of actions are down regulated so to speak by 100x before any evidence comes in then it's going to be interpreted within a prior that downregulates or basically downweights the relevance or the likelihood of certain types of actions happening but also there's this element of the preferences where it may through whatever mechanism it's not an exact metaphor here the system kind of like a Stockholm syndrome situation the system can actually uh uh internalize a preference for certain States that may or may not be consciously available during regular States Of Consciousness that lead into the implicit processing of everything within a framework that results in Behavior that's not aligned with something else which could be external Niche pressure or peer pressure or it could be discordant with one's own narrative themes and then meditation is interesting to think about as far as isolating these different types of parameters in the model and different kinds of meditation lots of different ways to do it Adam

hi

um hard to follow up on all that the um some the way the threads came together for me uh

was it seems like across varieties of meditation

and why would you do such a thing um and ultimately like to bring back maros

point about empowerment um it's you want

to be more capable in some way having a better grip on existence and a particular kind of way uh you want to be you have more you want to have um more degrees of freedom uh with respect to the way you're engaging with the world and there's a sense in which you can think of different types of meditation as either as allowing you to uh navigate through cybernetically uh free free energy Landscapes which can uh and and and also adjust the nature of these Landscapes such that they are navigable so uh you can think of something like and to bring back in like something like trauma or something like uh the elephant getting stuck you can think of this as like a a getting stuck at a local Optimum um of of a potentially Jagged free energy landscape and so if you can uh let's say reduce the uh some emotional uh regulation capacity the ability to defuse from an experience and keep the gain on positive feedback loop cycles of thought and emotion from getting as high as they might uh you might be able to make the free energy landscape less Jagged the differen is from point to point it's going to be uh you're not g to have these wild swings and so this might give you greater latitude with respect to pursuing more complex uh policies and actually I hope at some point we start to discuss some of colins's work on well-being because I think it it docks uh there's a lot to be said there um but uh one thing uh on

this one podcast con was mentioning like you can think of one of the benefits of mindfulness uh which is like in pursuing any kind of complex goal um there's going to be a sort of trading off you're going to have to do against different things on on road there and so if you can like temporarily uh let some of your characteristic adaptations slide um let some of your uh policies like not be as important if they're if they're not being you think they ought to be realized but they're not within some sort of like hierarchical hierarchical policy selection well then you can pursue uh you can hit targets you wouldn't otherwise you can you're a mountain car you can get to the top of the mountain um and the mountain car problem um so yeah nice cool Colin and then anyone else I just wanted to to say to point out that that's that's the essence of non-attachment right you're not attached to particular policies and that means that you don't get hung up on a subgoal when you should be shifting policies because what you are trying to pursue is some larger goal there's sort of you know there's The Stereotype of uh meditation at least you know Zen meditation for example Buddhist meditation that you're eliminating your goals right you have no desires but that's not that is a that's a caricature right because it's not that you have no goals it's just that you're not attached to particular low-level goals you might be uh you know you probably remain

attached to if you're a good Buddhist
for example to eliminating suffering for
all beings and so it's not like you're
not go directed and you know you
probably maintain maintain life by
eating and things like this it's just
that you're not attached to the
particular policies that might be using
uh to achieve these higher level goals
because goals are are just nested I mean
there's a way in which there's a goal
but then the policy is also just a goal
it's just there're nested goals all the
way down to the terminals which are
specific motor operations or specific
cognitive operations and so
non-attachment is basically becoming
more flexible in which subgoals you
are willing to consider and to
use very nice blue
so I wonder if it's like it's more that
it's not attachment to the policy but
that actually is the policy and it's not
attachment to the specific generative or
predictive model I wonder if that's like
because the policy is then the policy of
non-attachment to the model or or
Ultimate model
flexibility right you could call it a
metapolicy metapolicy right and and
so um and like secondly I wonder about
uh you know the idea of staying in the
in the present moment is this like some
kind of free energy equilibrium like in
and not necessarily minimization but
like if we take the generative model and
the predictive model if we take like the
value to zero are we like is meditation

like staying in the present moment just
like this ultimate state of of
equilibrium in in free energy like
infodynamics well interesting point
about the hyper prior the basian hyper
prior and also that's why a lot of
meditation processes involve rhythmic or
the cessation of other inputs because it
allows the interception potentially to
proceed in a different way isolation
tanks or uh other Altered States Of
Consciousness so let's do keeping
comment short because we only a few
minutes left Adam and then
Marco in terms of the there's this one
paper actually that just came out
recently about a model of meditation
with respect to um the temporal depth
and counterfactual richness of uh your
generative modeling that just came out
I'll try to find that and send that but
I think that would speak to this idea of
staying in the present moment also this
idea of um staying in the present moment
um also makes me think of this uh Mark
was like kind of intimating that you can
think of meditation as a kind of uh like
game task or almost like a kind of like
uh maybe maybe play even uh but it's
this
building
capacity um that
is building
capacity okay we'll come back Marco for
okay uh so so a building of capacity
through relaxing some of your priors and
so letting um you to be uh informed by
uh policies with respect to mental acts

that you might not normally deploy and you might not normally notice um what is entailed when you deploy certain types of mental act so things might come to mind uh that wouldn't normally uh that would place you now in into a different state of uh affective inference and so I think uh if we wanted to do uh modeling of this of something like meditation uh we would want to use this sort of imaginative policy selection described by something like sophisticated um effect of inference and that would begin to get us there um the exact way and one more thought before we move on is um one more thing I I was actually wondering with respect to meditation I've talked about this some with Colin is uh above the traits in the personality hierarchy you have these meta traits of stability and plasticity and um stability uh is this shared variance over and uh Colin actually named these um it's the shared variance over neuroticism inverse neuroticism agreeableness and conscientiousness and plasticity is the shared variance over extroversion and openness intellect and interestingly the stability seems to be basically the inverse of the general Psychopathology Factor P um by and large um but I I think thinking of mindfulness and meditation with resp back to um building capacities for both stability and plasticity which um helping to U um basically place this so you can think of stability as your ability to follow

through on and protect your policies or characteristic adaptations and then plasticity would be your tendency to form new characteristic adaptations so like whether um so let's say you do encounter trauma um plasticity might be what makes the difference so stability would be how much it might make sense to the degree to which you're actually traumatized like you're very stable something might you know bounce right off you you're you're not traumatized but let's say you don't have a lot of plasticity and that instability is compromised plasticity is the thing that would let you update and uh create new strategies uh for being in the world and so they're kind of the state of dynamic tension in that they can like pull at each other um but they're both ultimately synergistic in that like you're not going to be particularly plastic unless you have this Foundation of stability and you're not going to be able to maintain your stability long unless you're metastable and so um part of with meditation mindfulness I'm wondering if one way of thinking of it is it's part of what's actually really letting you kind of like ride this Optimum of balance stability and plasticity and that could be one way of thinking about part of what um people get from it cool Marco then Colin yeah that's amazing that's it's a perfect setup yeah um I I I also fully agree with this theme of complementarity

right so stability and plasticity sounds
really it resonates with picity sorry
openness and focus which is a big
complimentary Uhl yeah big po polarity
in meditation practices right so if open
meditation just feel everything let it
all come in ETC Focus meditation you
know focus on one point one
Mantra um and that's has a bigger
connection to the the more well-known
complimentarity or polarity of
exploration exploitation right but it's
not about the polarities per se it's
it's always about the balancing them Ma
and for example Sussex had also a nice
paper on that it's not about perfect
thetical representations you can have
some errors it's about how you balance
everything off and so again we go back
to the idea notion that that meditation
can be games where you effectively
practice and balancing that on cuz that
balance is not something you can
explicitly verbalize it's intrinsically
preverbal because you can verbalize
maybe the particular strategies or
beliefs for Ability or for picity but to
balance them off is more effective it's
more flexible or fluid and so the the
another interesting maybe Duality that
that you can think of is also
equilibriums and games or contexts right
so in Game Theory you have an
equilibrium for a particular game or
setting and so what you said earlier
about equilibrium and staying in the
moment uh regarding meditation I think
it's a very nice idea what what Adam

already said is that um by being in that particular game which is that meditative practice um you will naturally by free energy minimization go to a particular kind of equilibrium but that NE necessitates change some updates and those updates might actually mean that your equilibrium seeking Behavior or your attractors will be different in the other context so the underlying assumption is that these particular changes in equilibriums or priors or attractors um are transferable so um yeah does that make sense as for your question on meditation presence in the presence yeah cool thoughts

Colin I was just going to say a little bit more about stability and plasticity um I think that those can be seen as the two fundamental uh needs of any cybernetic or active inference system that exists in a complex or changing or unpredictable environment right so stability is in some sense just the the most fundamental requirement because the system must be able to pursue its goals effectively in order to be a cybernetic system right if it's an organism that's what it must have in order to persist and even an artificial system in order to persist as the system that it is it must be moving effectively toward its goals um but then if you have any degree of unpredictability in the environment you can't just be satisfied with the same set of policies that you have at any given time you're going to have to be able to update your policies generate

new policies and so then immediately in order to remain stable over time in a changing environment you also have to be somewhat adaptive or plastic and and so you have these uh inherent Necessities for the capacity to maintain a stable order of goal directed functioning and then the capacity to generate uh new components of the model right new policies or even new goals or new interpretations of the of the situation that you're in and then those so those are complimentary but they can also they are also in Dynamic tension as Adam mentioned because if you're massively exploratory it can be difficult to remain stable because you have so much new uh you know information coming in um similarly if you are well I mean we've essentially already covered what the the danger of being too inflexible and too unexplored as things change over time then you become unadapted to your situation so stability and plasticity aren't opposite to each other right stabil the opposite of stability is instability the opposite of plasticity is rigidity you can be uh stable and plastic or stable and rigid you can be unstable and plastic and unstable and rigid cool Adam go ahead

hi

um it seems like at some point it could be good to discuss connections of um narrative perspectives on personality because this idea of um narrative as active inference and the work trying to be done there I think there could be

like a rich interchange because in some ways you can think of know selves as centers of narrative gravity and you can think of person as um styles and modes of narration and themes and motifs as you are this semi-reliable narr narrator telling your story and making and enacting it and making it so to degrees um and this being a source of to the degree you have stability and plasticity um this would be one source of it the ability of narratives to provide cross situational intertemporal common context and but also because of their richness to create this um Rich this kind of their combinatorics and Rich representative capacity also being a source of plasticity and so um narrative being a source of balance Dynamic balancing this Dynamic tension although the narrative itself can also not be a source of balance but imbalance if the style of naration becomes problematic so that kind of mindfulness enters in there again where um it might like your usual styles of narrative narrative l ation you might let that go a bit you go um this is this is the way in which you go naive Bays on your expectations um yep the stories provide the polls and the archetypes and then you can say okay it's like Goldilocks but for the internet of things or something like that and that gives you a vocabulary to communicate and design you might need a very different story in a very different context and kind of bring it back to trauma that might be all the difference

between whether you get stuck or you experience post-traumatic growth yeah cool interesting and just um one closing uh slide that I want to put up while people prepare their thoughts this is a question just from uh audience member they wrote I have a question about what happens when blankets intersect blankets of blankets and so let's think about this in this kind of blankets of physiologically relevant States informationally relevant what what um think about what we're talking about here and at which nodes that would happen and then there's a link to a talk by Carl Friston and a question about how nodes intersect and just while people are thinking about this or their last thoughts on the paper uh one note is about the scientist and the personality scientist as an informational forager as an active inference agent themselves and so our quest to reduce uncertainty about personality it shouldn't be about the optimal descriptor it should lead to a question of ourselves which is what is the optimal perturbation that we can design that will reduce our uncertainty the most about this individual's personality and so that takes into account population level Dynamics as a prior but personalized and um kind of dynamic testing so imagine having a baseline prior across knowledge architectures that people had and then having a dynamically updating online course that was optimally informative from both

sides of the interface it was optimally
informative for what kind of learner the
person was and they were reducing their
uncertainty optimally about the topic at
hand and so that was like a
co-evolutionary process
amidst a prior that was population level
but personalized in the individual
trajectory so it's a really interesting
framework in col and Adam thanks a lot
for coming on the stream was like really
helpful and we're excited that you'll
come back for next week so in these last
couple minutes anyone can raise your
hand give a last thought what would be
something that we would want to say to
this slide about how nodes intersecting
stories intersecting individuals
intersecting how does that come into
play and or what should we be thinking
about for 13.2 that would be good to
cover Shannon and then anyone else who
raises their
hand I guess I have a thought about um
the question on this slide um so is um
one node that's a sent state in one
blanket the internal state of another
blanket like that's totally the case
because you basically choose a
perspective and whichever you know
blanket sort of internal State you want
to describe or whichever thing that
you're calling the agent separate from
its environment then that's the internal
state that you're going to describe and
if you're looking at like if this blue
node is a neuron and you want to
actually know about this pink node this

other neuron's internal State then
that's going to be your internal State
and these other guys are going to be the
sent State the blanket around it and
that's going to be the statistical
boundary that you're using to explain
the behavior of that neuron and then you
can Aggregate and this collection of
neurons will all be an internal State
for the next collection of neurons so
the answer is in the question that's
exactly right perfect answer showing
that the answer is in the question is
the answer

Adam uh in terms of uh loop around to a
previous thought in terms of nested uh
blankets um and where these different
blankets might have different
characteristics in their interaction
um for personality uh there's some
interesting work um colum might disagree
on tight and loose cultures where you
can think of like the personality of the
culture you're within and this tightness
and looseness seems to correspond to
this stability plasticity Dynamic
tension this fundamental cybernetic
parameters showing up on the group level
and you in theory can get this kind of
complimentarity there where a tight
culture might create more tight um
individuals with respect to the degree
to which you they let their their Pol
whether they have a metap policy of
letting their policy slide more um in
different context but uh or instead of
it being this sort of entrainment
relation it could be a complimentary

relation where the stability on the level of the group this broader um uh attractor affords the fine grain the subsystems to actually have more latitude and vice versa um and I I imagine you might probably see similar things play out like with ant colonies um interesting point about multi-level systems and how they're associated with relative constraints in the last two three minutes let's think of a last you know if anyone who wants to raise their hand give a word or a sentence or few what is something that you took away or was exciting today and what was something that we could continue talking about in 13.2 and pick up where we left off Marco and then just anyone else who raises their hand yeah I just wanted to thank everyone this was like really amazing this my first this this kind of stream seminar uh in ages so thank you is innovating and I think in general what to be taken away I think is is uh excitement because it's kind ofing how many different domains and topics and questions have been successfully synthesized in this big discussion and I think that's a testament to the power of these kinds of Frameworks of invasion or Active influence frean Energy C etics when it comes to understanding us people us persons personalities um as individuals as as collectives so thank you again for exciting me thank you well said and the group performance is uh only realized it's only in acted even

digitally when the people show up and it's only relevant when there's the regime of attention and so there's so much you know fun and it's great to have you um multiple of you on your first time here that's always appreciated so any other thoughts in the last minute otherwise in the calendar invite is a feedback form so if you're just wanting to think a little bit more if you're live participant you want to think a little more about what we could cover next week just put it in that feedback form and then anyone who's watching it just leave a comment for something that we could address even if it's after 13.2 maybe we can swoop back and return to some previous comments but this was a super fun discussion and a big thanks to all the participants so really good times everyone and thanks a lot so see you next week